27 Dual Nature of Radiation and Matter

Frequency

1. If a photon has velocity c and frequency v, which of the following represents its wavelength?
   (a) $\frac{hv}{c^2}$ (b) hv (c) $\frac{hc}{v}$ (d) $\frac{c}{v}$
2. Find frequency of 1 MeV Photon?
3. The velocity of photons is proportional to (where υ = frequency)
   (a) $\frac{1}{\sqrt{\upsilon}}$ (b) υ^2 (c) υ^0 (d) $\sqrt{\upsilon}$

Energy of Photon

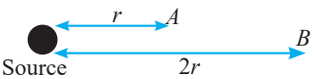
4. Energy of a quanta of frequency 10^{15} Hz and $h = 6.6 \times 10^{-34}$ Js will be:
   (a) 6.6×10^{-19} J (b) 6.6×10^{-12} J
   (c) 6.6×10^{-49} J (d) 6.6×10^{-41} J
5. If radiation of energy E and number of photon n then find effective moving mass of photon.
   (a) $\frac{E}{C}$ (b) $\frac{E}{C^2}$ (c) $\frac{E}{nC^2}$ (d) $\frac{nE}{C^2}$
6. A photon in motion has a mass:
   (a) $\frac{c}{hv}$ (b) $\frac{h}{v}$ (c) hv (d) $\frac{hv}{c^2}$

Power of Light

7. If Green light and Red light having same power then which have large number of photon.
   (a) $n_R = n_G$ (b) $(n_R > n_G)$
   (c) $n_R < n_G$ (d) Can't say
8. If power of light 10 kW and wavelength 300 nm then find number of photon per sec?
9. Two sources of light emit with a power of 200 W. The ratio of number of photons of visible light emitted by each source having wavelengths 300 nm and 500 nm respectively, will be:
   (a) 1 : 5 (b) 1 : 3 (c) 5 : 3 (d) 3 : 5
10. Monochromatic light of frequency 6×10^{14} Hz is produced by a laser. The power emitted is 2×10^{-3} W. How many photons per second on an average, are emitted by source?
    (Given $h = 6.63 \times 10^{-34}$ Js)
    (a) 9×10^{18} (b) 6×10^{15} (c) 5×10^{15} (d) 7×10^{16}

Intensity of Light Beam

11. If intensity becomes double then number of photon will be?
    (a) Same (b) Double (c) Half (d) Four time
12. If frequency and intensity both becomes double then number of photon will be.
    (a) Same (b) Double (c) Half (d) Four time
13. Find: (i) $\left(\frac{f_A}{f_B}\right)_{\text{Photon}}$ (ii) $\left(\frac{I_A}{I_B}\right)_{\text{Light}}$



    (iii) Ratio of number of Photon per unit Area
14. Consider a parallel beam of light of wavelength 600 nm and intensity 100 W/m^2. How many photons cross 1 cm^2 area perpendicular to beam in one second?
    (a) 3×10^{16} (b) 3×10^{14} (c) 3×10^{9} (d) 3×10^{4}
15. Monochromatic light of wavelength 3000 Å is incident on a surface area 4 cm^2. If intensity of light is 150 W/m^2, find the rate at which photons strike the target.

Radiation Pressure

16. If intensity of light is I and 75% radiation is absorbed then find radiation pressure?
17. Radiation pressure in terms of absorption coefficient and Reflection coefficient?
18. A plate of Mass 10 g is in equilibrium in air due to light bean on plate then power of beam, if surface area of plates is A and complete reflecting.

Momentum

19. The momentum of the photon of wavelength 5000 Å will be:

(*a*) 1.3×10^{-27} kg-m/s (*b*) 1.3×10^{-28} kg-m/s

(*c*) 4×10^{-29} kg-m/s (*d*) 4×10^{-18} kg-m/s

20. If the total energy transferred to a surface in time t is 6.48×10^5 J, then the magnitude of the total momentum delivered to this surface for complete absorption will be:

(*a*) 2.46×10^{-3} kg-m/s (*b*) 2.16×10^{-3} kg-m/s

(*c*) 1.58×10^{-3} kg-m/s (*d*) 4.32×10^{-3} kg-m/s

21. Given below are two statements:

Statement-I: Two photons having equal linear momenta have equal wavelengths.

Statement-II: If the wavelength of photon is decreased, then the momentum and energy of a photon will also decease.

Statement-III: Photon having same wavelength have equal momentum

(*a*) TFF (*b*) FTF (*c*) FFT (*d*) FFF

Photoelectric Effect

22. Which one among the following shows particle nature of light?

(*a*) Photoelectric effect

(*b*) Interference (Wave)

(*c*) Refraction (Wave)

(*d*) Polarization (Transverse wave)

23. In photo electric effect:

A. The photocurrent is proportional to the intensity of the incident radiation.

B. Maximum kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.

C. Max. K.E with which photoelectrons are emitted depends on the frequency of incident light.

D. The emission of photoelectrons require a minimum threshold intensity of incident radiation.

E. Max. K.E of the photoelectrons is independent of the frequency of the incident light.

Choose the correct answer from the options given below:

(*a*) A and C only (*b*) A and E only

(*c*) B and C only (*d*) A and B only

24. When light of wavelength 300 nm falls on a photoelectric emitter, photoelectrons are just liberated. For another emitter, light of wavelength 600 nm is just sufficient for liberating photoelectrons. The ratio of the work function of the two emitters is:

(*a*) 1 : 2 (*b*) 2 : 1

(*c*) 4 : 1 (*d*) 1 : 4

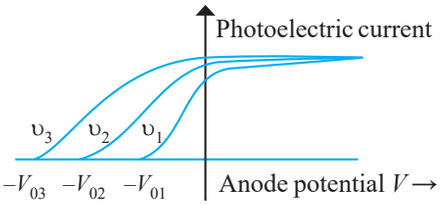
25. A 2 mW laser operates at a wavelength of 500 nm. The number of photons that will be emitted per second is:

(Given Planck's constant $h = 6.6 \times 10^{-34}$ Js)

(*a*) 2×10^{16} (*b*) 1.5×10^{16}

(*c*) 5×10^{15} (*d*) 1×10^{16}

26. In photoelectric effect, the curve between photoelectric current and anode potential V (for different frequencies) is shown in figure, then:



(*a*) $\upsilon_1 > \upsilon_2 > \upsilon_3$ (*b*) $\upsilon_1 < \upsilon_2 < \upsilon_3$

(*c*) $\upsilon_1 = \upsilon_2 = \upsilon_3$ (*d*) $\upsilon_1 > \upsilon_2 < \upsilon_3$

27. Light of frequency 1.5 times the threshold frequency is incident on photo-sensitive material. If the frequency is halved and intensity is doubled, the photo current becomes:

(*a*) Quadrupled (*b*) Doubled

(*c*) Halved (*d*) Zero

28. For a certain metal, v is five times the v_0 and the maximum velocity of coming out photo electrons is 8×10^6. If $v = 2v_0$ the maximum velocity of photo electrons will be:

(*a*) 4×10^6 m/s (*b*) 6×10^6 m/s

(*c*) 8×10^6 m/s (*d*) 1×10^6 m/s

29. From the photoelectric effect experiment, following observations are made. Identify which of these are correct.

1. The stopping potential depends only on the work function of the metal.
2. The saturation current increases as the intensity of incident light increases.
3. The maximum kinetic energy of a photo electron depends on the intensity of the incident light.
4. Photoelectric effect can be explained using wave theory of light.

Choose the correct answer from the options given below:

(*a*) 2, 3 only (*b*) 1, 3, 4 only

(*c*) 2 only (*d*) 1, 2, 4 only

30. A metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V_0. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential becomes $V_0/4$. The threshold wavelength for this metallic surface will be:

(*a*) $\frac{\lambda}{4}$ (*b*) 4λ (*c*) $\frac{3}{2\lambda}$ (*d*) 3λ

31. When photon of energy 3.8 eV falls on metallic surface of work function 2.8 eV, then the kinetic energy of emitted electrons are:

(*a*) 1 eV (*b*) 6.6 eV (*c*) 0 to 1 eV (*d*) 2.8 eV

32. A monochromatic point source of light is placed at a distance d from a metal surface. Photo electrons are ejected at a rate n per second, and with maximum kinetic energy E. If the source is brought nearer to distance $d/2$, the rate and the maximum kinetic energy per photoelectron become nearly.

(a) $2n$ and $2E$ (b) $4n$ and $4E$
(c) $4n$ and E (d) n and $4E$

33. The photoelectric work function:

(a) Is different for different materials.
(b) Is same for all metals.
(c) Depends upon frequency of the incident light
(d) Depends upon intensity of the incident light

34. Two identical photocathodes receive the light of frequencies f_1 and f_2 respectively. If the velocities of the photo-electrons coming out are v_1 and v_2 respectively, then:

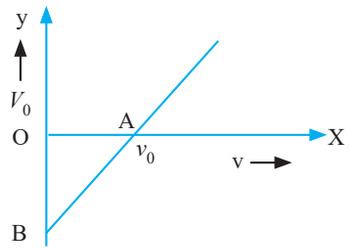
(a) $v_1^2 - v_2^2 = \frac{2h}{m}[f_1 - f_2]$ (b) $v_1^2 + v_2^2 = \frac{2h}{m}[f_1 + f_2]$

(c) $v_1 + v_2 = \left[\frac{2h}{m}(f_1 + f_2)\right]^{\frac{1}{2}}$ (d) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{\frac{1}{2}}$

35. If the frequency of light incident on a metallic plate the doubled, how will the maximum kinetic energy of the photoelectrons change?

(a) It becomes more than double
(b) It becomes less than double
(c) It becomes exactly double
(d) It does not change

36. In an experiment on photoelectric effect the frequency f of the incident light is plotted against the stopping potential V_0. The work function of the photoelectric surface is given by (e is electronic charge).



(a) $OB \times e$ in eV (b) OB in volt
(c) OA in eV (d) The slope of the line AB

37. Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be:

(a) 1 : 2 (b) 1 : 3 (c) 2 : 3 (d) 3 : 2

38. In photoelectric effect, the slope of stopping potential versus frequency of incident light for a given surface will be:

(a) $h\,e^{-1}$ (b) $e\,h$ (c) e (d) h

39. Wave nature of light cannot explain photoelectric effect because in photoelectric effect, it is seen that:

(a) For the frequency of light below a certain value, the photoelectric effect does not take place, irrespective of intensity.
(b) Maximum kinetic energy of ejected electrons is independent of intensity of radiation.
(c) There is no time lag between the incidence of radiation and emission of electrons.
(d) All of these

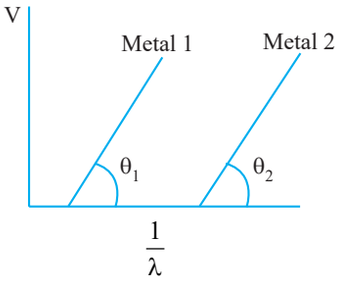
40. The work function of a substance is 4eV. The longest wavelength of light that can cause the emission of photoelectrons from this substance is approximately.

(a) 540 nm (b) 400 nm (c) 310 nm (d) 220 nm

41. The work functions of Aluminium and Gold are 4.1 eV and 5.1 eV respectively. The ratio of the slope of the stopping potential versus frequency plot for Gold to that of Aluminium is:

(a) 1.24 (b) 2 (c) 1 (d) 1.5

42. V (stopping potential) is plotted against 1/λ, where λ is wavelength of incident radiations, for two metals:



(a) Metal 1 may be gold and metal 2 may be cesium
(b) $\theta_1 > \theta_2$, if metal-1 is gold and metal-2 is cesium
(c) $\theta_1 = \theta_2$, for any two metals
(d) $\theta_1 > \theta_2$, if metal-1 and metal-2 are gold and copper respectively.

43. The cathode of photocell is changed such that the work function changes from w_1 ($w_2 > w_1$). If the saturation currents before and after the change are I_1 and I_2 and all other conditions are unchanged, then (assuming $h\nu > w_2$).

(a) $I_1 = I_2$ (b) $I_1 < I_2$
(c) $I_1 > I_2$ (d) $I_1 < I_2 > 2I_2$

44. The difference between threshold wavelength for two metal surfaces A and B having work function $\phi_A = 9$ eV and $\phi_B = 4.5$ eV in nm is:

(a) 264 (b) 138 (c) 276 (d) 540

45. If the two metals A and B are exposed to radiation of wavelength 350 nm. The work functions of metals A and B are 4.8 eV and 2.2 eV. Then choose the correct option.

(a) Metal B will not emit photo-electrons.
(b) Both metals A and B will emit photo-electrons.
(c) Both metals A and B will not emit photoelectrons.
(d) Metal A will not emit photo-electrons.

46. The threshold frequency of metals is f_0. When the light of frequency $2f_0$ is incident on the metal plate, the maximum velocity of photoelectron is v_1. When the frequency of incident radiation is increased to $5f_0$. The maximum velocity of photoelectrons emitted is v_2. The rato of v_1 to v_2 is:

(*a*) $\frac{v_1}{v_2}=\frac{1}{2}$ (*b*) $\frac{v_1}{v_2}=\frac{1}{8}$ (*c*) $\frac{v_1}{v_2}=\frac{1}{16}$ (*d*) $\frac{v_1}{v_2}=\frac{1}{4}$

47. Given below are two statements:

Assertion (A): The photoelectric effect does not takes place, if the energy of the incident radiation is less than the work function of metal.

Reason (R): Kinetic energy of the photoelectrons is zero, if the energy of the incident radiation is equal to the work function of a metal.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are correct and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are correct but Reason (R) is the correct explanation of Assertion (A).

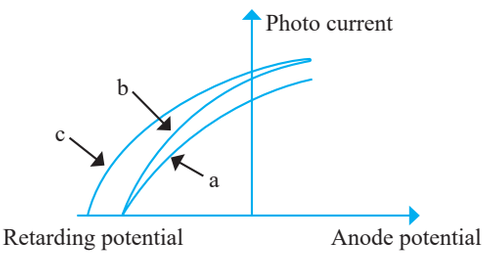
(*c*) Assertion (A) is correct but Reason (R) is not correct.

(*d*) Assertion (A) is not correct but Reason (R) is correct.

48. The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ To double the kinetic energy, the incident light must have wavelength:

(*a*) $\frac{hc}{E\lambda-hc}$ (*b*) $\frac{hc\lambda}{E\lambda+hc}$ (*c*) $\frac{h\lambda}{E\lambda+hc}$ (*d*) $\frac{hc\lambda}{E\lambda-hc}$

49. The figure shows a plot of photo current versus anode potential for a photosensitive surface for three different radiations. Which one of the following is a correct statement?



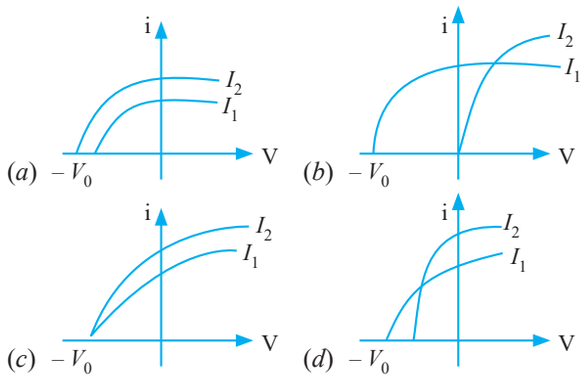
(*a*) Curves (*b*) and (*c*) represent incident radiations of same frequency having same intensity.

(*b*) Curves (*a*) and (*b*) represent incident radiations of different frequencies and different intensities.

(*c*) Curves (*a*) and (*b*) represent incident radiations of same frequency but of different intensities.

(*d*) Curves (*b*) and (*c*) represent incident radiations of different frequencies and different intensities.

50. Which figure shows the correct variation of applied potential difference (V) with photoelectric current (I) at two different intensities of light ($I_1 < I_2$) of same wavelengths:



51. As the intensity of incident light increases:

(*a*) kinetic energy of emitted photoelectrons increases.

(*b*) photoelectric current decreases.

(*c*) photoelectric current increases.

(*d*) kinetic energy of emitted photoelectrons decreases.

52. A photocell employs photoelectric effect to convert:

(*a*) change in the frequency of light into a change in the electric current.

(*b*) change in the frequency of light into a change in electric voltage.

(*c*) change in the intensity of illumination into a change in photoelectric current.

(*d*) change in the intensity of illumination into a change in the work function of the photocathode.

53. In a photoelectric experiment, increasing the intensity of incident light:

(*a*) Increases the number of photons incident and also increases the K.E of the ejected electrons.

(*b*) Increases the number of photons incident and the K.E. of the ejected electrons remains unchanged.

(*c*) Increases the frequency of photons incident and increases the K.E of the ejected electrons.

(*d*) Increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged.

54. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $V/4$. The threshold wavelength for the metallic surface is:

(*a*) 5/2 λ (*b*) 3 λ (*c*) 4 λ (*d*) 5 λ

55. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and λ/2. If the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case, the work function of the surface of the material is (h = Planck's constant, c = speed of light).

(*a*) $2hc/\lambda$ (*b*) $hc/3\lambda$ (*c*) $hc/2\lambda$ (*d*) hc/λ

56. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Two photons each of energy 3 eV, can't eject electrons from a surface of work function 5 eV.

Reason (R): Only a photon of energy greater than work function may emit photoelectron.

In the light of the above statements, choose the most appropriate answer from the options given below:

(*a*) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(*b*) Both Assertion (A) and Reason (R) are true and Reason (R) is Not the correct explanation of Assertion (A).

(*c*) Assertion (A) is true but Reason (R) is false.

(*d*) Assertion (A) is false but Reason (R) is true.

57. On a photosensitive surface of threshold wavelength 6λ, two different monochromatic beams of wavelengths 4λ and 2λ are incident, one after another, then match the ratio of physical quantities specified in column-I to the ratio values given in column-II.

Column-I		Column-II	
(i)	Maximum KE of photo electrons	A.	1 : 2
(ii)	Speed of incident photons	B.	1 : 4
(iii)	Momentum of incident photons	C.	1 : 2
(iv)	Fastest speeds of photo electrons	D.	1 : 1

Choose the correct answer from the option given below:

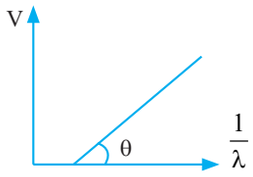
(*a*) i - C, ii - B, iii - A, iv - D

(*b*) i - B, ii - D, iii - C, iv - A

(*c*) i - B, ii - D, iii - A, iv - C

(*d*) i - A, ii - B, iii - D, iv - C

58. In photoelectric effect experiment, the graph of stopping potential V versus reciprocal of wavelength obtained is shown in the figure. As the intensity of incident radiation is increased:



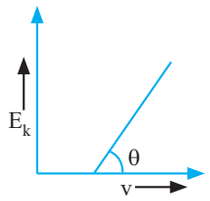
(*a*) Slope of the straight line get more steep

(*b*) Straight line shifts to left

(*c*) Straight line shifts to right

(*d*) Graph does not change

59. For the photoelectric effect, the maximum kinetic energy (E_k) of the photoelectrons is plotted against the frequency (v) of the incident photons as shown in figure. The slope of the graph gives:



(*a*) Ratio of Planck's constant to electric charge.

(*b*) Work function of the metal

(*c*) Charge of electron

(*d*) Planck's constant

60. Which of the following statement is not true about stopping potential (V_0)?

(*a*) It depends on the nature of emitter material.

(*b*) It depends upon frequency of the incident light.

(*c*) It increases with increase in intensity of the incident light.

(*d*) It is $1/e$ times the maximum kinetic energy of electrons emitted.

61. In photoelectric experiment energy of 2.48 eV irradiates a photo sensitive material. The stopping potential was measured to be 0.5 V. Work function of the photo sensitive material is:

(*a*) 0.5 eV (*b*) 1.68 eV (*c*) 2.48 eV (*d*) 1.98 eV

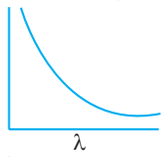
62. The work function of a substance is 3.0 eV. The longest wavelength of light that can cause the emission of photoelectrons from this substance is approximately:

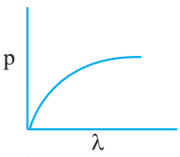
(*a*) 215 nm (*b*) 414 nm (*c*) 400 nm (*d*) 200 nm

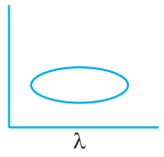
De - Broglie Wavelength

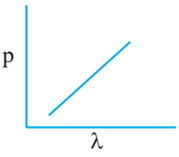
63. Find the ratio of velocities of proton and α-particle if the de-Broglie wavelengths of both the particles is same.

64. Variation of momentum of particle (p) with associated de-Broglie wavelength (λ) is shown correctly by:

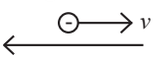
(*a*) 

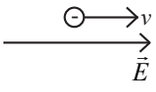
(*b*) 

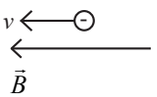
(*c*) 

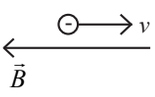
(*d*) 

65. Figure shows four situations in which an electron is moving in electric / magnetic field. In which case the de-Broglie wavelength of electron is increasing?

(*a*) 

(*b*) 

(*c*) 

(*d*) 

66. The de-Broglie wavelength of a particle accelerated with 150 volt potential is 10^{-10} m. Find the wavelength of the particle if it is accelerated by 600 volt potential difference.

67. The de-Broglie wavelength of a neutron at 27°C is λ. What will be it wavelength at 927°C?

(*a*) λ/2 (*b*) λ/3 (*c*) λ/4 (*d*) λ/9

68. An electron of mass *m*, when accelerated through a potential difference *V*, has de-Broglie wavelength λ. The de-Broglie wavelength associated with a proton of mass *M* accelerated through the same potential difference, will be.

(*a*) $\lambda\frac{M}{m}$ (*b*) $\lambda\frac{m}{M}$ (*c*) $\lambda\sqrt{\frac{M}{m}}$ (*d*) $\lambda\sqrt{\frac{m}{M}}$

69. If the velocity of an electron is doubled, its de-Broglie frequency will:

(*a*) Be halved (*b*) Remain same

(*c*) Be doubled (*d*) Become four times

70. An α-particle moves in a circular path of radius 0.83 cm in the presence of a magnetic field of 0.25 Wb/m^2. The de-Broglie wavelength associated with the particle will be:

(*a*) 10 Å (*b*) 0.01 Å (*c*) 1 Å (*d*) 0.1 Å

71. De-Broglie wavelength associated with an electron revolving in the nth state of hydrogen atom is directly proportional to:

(*a*) n (*b*) $1/n$ (*c*) n^2 (*d*) $1/n^2$

72. A proton is accelerated through 225 *V*. Its de-Broglie wavelength is:

(*a*) 0.1 nm (*b*) 0.2 nm (*c*) 0.3 nm (*d*) 0.4 nm

73. An electron of mass *m* and a photon have same energy *E*. The ratio of de-Broglie wavelengths associated with them is:

(*a*) $c(2mE)^{\frac{1}{2}}$ (*b*) $\frac{1}{c}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$ (*c*) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (*d*) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

74. The wavelength λ_e of an electron and λ_p of a photon of same energy *E* are related by:

(*a*) $\lambda_p \propto \sqrt{\lambda_e}$ (*b*) $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$ (*c*) $\lambda_p \propto \lambda_e^2$ (*d*) $\lambda_p \propto \lambda_e$

75. Of the following having the same kinetic energy, the one which has the largest wavelength is:

(*a*) An alpha particle (*b*) A neutron

(*c*) A proton (*d*) An electron

76. The de-Broglie wavelength λ associated with a proton increases by 25%, if its momentum is decreased by p_0. The initial momentum was:

(*a*) $4p_0$ (*b*) $\frac{p_0}{4}$ (*c*) $5p_0$ (*d*) $\frac{p_0}{5}$

77. The de-Broglie wavelength of an electron in the 3rd Bohr orbit is related to the radius *R* of the orbit as:

(*a*) $3\lambda = \pi R$ (*b*) $\lambda = \frac{3}{2}\pi R$ (*c*) $3\lambda = 2\pi R$ (*d*) $3\lambda = 4\pi R$

78. When the momentum of a proton is changed by an amount p_0 the corresponding change in the de-Broglie wavelength is found to be 0.25%. Find the original momentum of the photon.

79. An electron (mass *m*) with an initial velocity $\vec{v} = v_0\hat{i}$ ($v_0 > 0$) is moving in an electronic field $\vec{E} = -E_0\hat{i}$ ($E_0 > 0$) where E_0 is constant. if at $t = 0$ de-Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de-Broglie wavelength after time *t* is given by:

(*a*) λ_0 (*b*) $\lambda_0\left(1+\frac{eE_0t}{mv_0}\right)$

(*c*) $\lambda_0 t$ (*d*) $\frac{\lambda_0}{\left(1+\frac{eE_0t}{mv_0}\right)}$

80. An electron of mass *m* and magnitude of charge | *e* | initially at rest gets accelerated by a constant electric field *E*. The rate of change of de-Broglie wavelength of this electron at time *t* ignoring relativistic effects is:

(*a*) $-\frac{h}{|e|Et}$ (*b*) $-\frac{h}{|e|E\sqrt{t}}$ (*c*) $\frac{-h}{|e|Et^2}$ (*d*) $\frac{|e|Et}{h}$

81. A particle of mass *M* is at rest, and exploid into two mass of ratio 2 : 3 then find ratio of de-Broglie wavelength of the two mass:

(*a*) 2 : 3 (*b*) 3 : 2 (*c*) 1 : 1 (*d*) 4 : 9

82. Proton, deutron, α-particle are moving with same momentum (*m, e*), (2*m, e*), (4*m,* 2*e*) then find ratio of their de-Broglie wavelength?

83. Proton, deutron, α-particle are moving with same speed (*m, e*), (2*m, e*), (4*m,* 2*e*) then find ratio of their de-Broglie wavelength?

84. Proton, deutron, α-particle are moving with same speed *K.E.* (*m, e*), (2*m, e*), (4*m,* 2*e*) then find ratio of their de-Broglie wavelength?

85. Proton, deutron, α-particle are accelerated with same potential difference then find ratio of de-Broglie?

86. De-Broglie wavelength of a body of mass 1 kg moving with velocity of 2000 m/s is:

(*a*) 3.32×10^{-27} Å (*b*) 1.5×10^{7} Å

(*c*) 0.55×10^{-22} Å (*d*) 2×10^{20} Å

87. The de-Broglie wavelength of an electron and the wavelength of a photon are the same. The ratio between the energy of that photon and the momentum of that electron is:

(*a*) h (*b*) c (*c*) $\frac{1}{h}$ (*d*) $\frac{1}{c}$

88. Which one of the following statements is not true about de-Broglie waves?

(*a*) All atomic particles in motion have wave character associated with them.

(*b*) The higher the momentum, the longer is the wavelength.

(*c*) The faster the particle, the shorter is the wavelength.

(*d*) For the same velocity, a heavier particle has a shorter wavelength.

89. A nucleus A, with a finite de-Broglie wavelength λ_A, undergoes spontaneous fission into two nuclie B and C of equal mass. B flies in the same direction as that of A, while C flies in the opposite direction with a velocity equal to half of that of B. The de-Broglie wavelengths λ_B and λ_C of B and C are respectively:

(*a*) $2\lambda_A, \lambda_A$ (*b*) $\lambda_A, 2\lambda_A$ (*c*) $\lambda_A, \frac{\lambda_A}{2}$ (*d*) $\frac{\lambda_A}{2}, \lambda_A$

90. Two particles move at right angle to each other. Their de-Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de-Broglie wavelength λ, of the final particle, is given by:

(*a*) $\lambda = \frac{\lambda_1 + \lambda_2}{2}$ (*b*) $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

(*c*) $\lambda = \sqrt{\lambda_1 \lambda_2}$ (*d*) $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

91. An electron moving with speed v and a photon moving with speed c, have same d-Broglie wavelength. The ratio of kinetic energy of electron to that of photon is:

(*a*) $\frac{v}{2c}$ (*b*) $\frac{2c}{v}$ (*c*) $\frac{v}{3c}$ (*d*) $\frac{3c}{v}$

92. The kinetic energy of an electron, α-particle and a proton are given as $4K$, $2K$ and K respectively. The de-Broglie wavelength associated with electron (λ_e) α-particle (λ_α) and the proton (λ_p) are as follows:

(*a*) $\lambda_\alpha = \lambda_p < \lambda_e$ (*b*) $\lambda_\alpha > \lambda_p < \lambda_e$

(*c*) $\lambda_\alpha < \lambda_p < \lambda_e$ (*d*) $\lambda_\alpha = \lambda_p < \lambda_e$

93. Electron beam used in an electron microscope, accelerated by a voltage of 20kV has a de-Broglie wavelength of λ_0. If the voltage is increased to 40kV then the de-Broglie wavelength associated with the electron beam would be:

(*a*) $3\lambda_0$ (*b*) $9\lambda_0$

(*c*) $\frac{\lambda_0}{2}$ (*d*) $\frac{\lambda_0}{\sqrt{2}}$

94. The de-Broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of K.E. of electron and K.E. of photon will be:

(*a*) $\frac{1}{1}$ (*b*) $\frac{1}{8}$

(*c*) $\frac{8}{1}$ (*d*) $\frac{1}{4}$

95. The de-Broglie wavelengths of a proton and an α-particle are λ and 2λ respectively. The ratio of the velocities of proton and α-particle will be:

(*a*) 1 : 8 (*b*) 1 : 2

(*c*) 4 : 1 (*d*) 8 : 1

MR* CORNER

1. **Statement I:** Two photons having equal frequencies, will have equal linear momenta.

   Statement II: Two photons having equal wavelengths, will have equal linear momenta.

2. Let P and E denote the linear momentum and energy of a photon. If the wavelength is decreased, P will____ whereas E will____

3. **Statement I:** The number of photons emitted by a red bulb and a blue bulb of equal power in a given time will be equal.

   Statement II: The equation E = pc is valid for a photon but not for an electron.

4. **Statement I:** When stopping potential is applied in an experiment on photoelectric effect, no photocurrent is observed. This means that the emission of photoelectrons is stopped.

   Statement II: Light of wavelength λ falls on metal having work function hc/λ_o. Photoelectric effect will take place only $\lambda \geq \lambda_o$

5. **Statement I:** The frequency and intensity of light source are both doubled. Then saturation photocurrent remains almost the same.

   Statement II: For above given case, the stopping potential will Become more than double.

6. **Statement I:** A non-monochromatic light is used in an experiment on photoelectric effect. The stopping potential will be related to the longest wavelength.

   Statement II: The De-Broglie wavelength of electron will be greater as compared to that of a proton if they are accelerated by the Same potential difference.

7. **Statement I:** When the intensity of a light source is increased, the Total Energy of the photons emitted per unit time increases.

   Statement II: Photoelectric effect supports quantum nature of light because electric charge of the photoelectrons is quantized.

8. **Statement I:** A photon of energy $h\nu$ is absorbed by a free electron of a metal having work function $\phi < h\nu$, then the electron is sure to come out.

   Statement II: For above given case, the electron is sure to come out with a kinetic energy of $h\nu - \phi$.

9. **Statement I:** If the wavelength of light in an experiment on photoelectric effect is doubled, the stopping potential will decrease.

   Statement II: The photocurrent in an experiment on photoelectric effect increases if the exposure time is increased.

10. **Statement I:** The collector plate in experiment on photoelectric effect is kept vertically above the emitter plate. If electric field is switch on which has a vertically downward direction, then the kinetic energy of electrons will increase.

    Statement II: In case of two particles fallen through same height, the heavier of two particles will possess smaller de Broglie wavelength.

Answer Key

1. (*d*) **2.** 2.4×10^{20}Hz **3.** (*c*) **4.** (*a*) **5.** (*c*) **6.** (*d*) **7.** (*b*) **8.** 15×10^{21} **9.** (*d*)

10. (*c*) **11.** (*b*) **12.** (*a*) **13.** (i) 1 : 1 , (ii) 4 : 1, (iii) 4 : 1 **14.** (*a*) **15.** 9×10^{13} sec^{-1} **16.** $\frac{5I}{4c}$

17. $P = \frac{I}{C}(\text{Absorption}),\ P = \frac{2I}{C}(\text{Reflection})$ **18.** 15 MW **19.** (*a*) **20.** (*b*) **21.** (*a*) **22.** (*a*) **23.** (*a*)

24. (*b*) **25.** (*c*) **26.** (*b*) **27.** (*d*) **28.** (*a*) **29.** (*c*) **30.** (*d*) **31.** (*c*) **32.** (*c*) **33.** (*a*)

34. (*a*) **35.** (*a*) **36.** (*a*) **37.** (*c*) **38.** (*a*) **39.** (*d*) **40.** (*c*) **41.** (*c*) **42.** (*c*) **43.** (*a*)

44. (*b*) **45.** (*d*) **46.** (*a*) **47.** (*b*) **48.** (*b*) **49.** (*c*) **50.** (*c*) **51.** (*c*) **52.** (*c*) **53.** (*b*)

54. (*b*) **55.** (*c*) **56.** (*a*) **57.** (*c*) **58.** (*d*) **59.** (*a*) **60.** (*c*) **61.** (*d*) **62.** (*b*) **63.** 4 : 1

64. (*a*) **65.** (*b*) **66.** **0**.5 $\times 10^{-10}$ m **67.** (*a*) **68.** (*d*) **69.** (*c*) **70.** (*b*) **71.** (*a*) **72.** (*d*)

73. (*c*) **74.** (*c*) **75.** (*d*) **76.** (*c*) **77.** (*c*) **78.** $400p_0$ **79.** (*d*) **80.** (*c*) **81.** (*c*)

82. (1 : 1 : 1) **83.** (4 : 2 : 1) **84.** 2 : $\sqrt{2}$: 1 **85.** 2 $\sqrt{2}$: 2 : 1 **86.** (*a*) **87.** (*b*) **88.** (*b*) **89.** (*d*)

90. (*d*) **91.** (*a*) **92.** (*c*) **93.** (*d*) **94.** (*b*) **95.** (*d*)

MR* CORNER

1. Both False **2.** Increase, Increase. **3.** I. False, II. True **4.** I. Both False **5.** Both are True.
6. I. False, II. True **7.** I. True, II. False **8.** Both False **9.** I. True, II. False **10.** Both True